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Tutorial Course for Intro to Sensor-Integrated Systems

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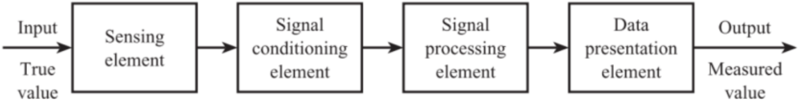
The General Measurement System

The **accuracy** of the system can be defined as the closeness of the measured value to the true value. A perfectly accurate system is a theoretical ideal and the accuracy of a real system is quantified using **measurement system error** E, where

$$E = \text{measured value} - \text{true value}$$

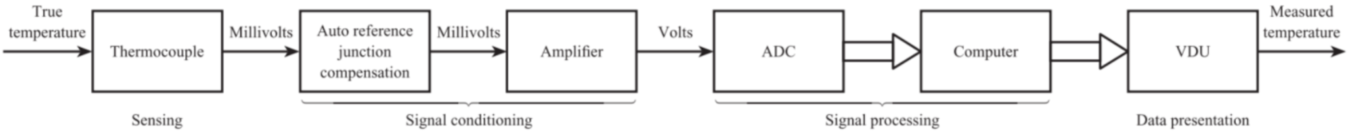
$$E = \text{system output} - \text{system input}$$

Structure of measurement systems



Examples of measurement systems

(a)



Static Characteristics of Measurement System Elements

Range specifies the minimum and maximum values of an input or output variable.

Input range:

$$I_{min} \rightarrow I_{max}$$

Output range:

$$O_{min} \rightarrow O_{max}$$

Range describes the allowable limits of operation. It does **not** indicate accuracy or precision.

Span is the difference between the maximum and minimum values.

Input span:

$$\text{Input } Span = I_{max} - I_{min}$$

Output span:

$$\text{Output } Span = O_{max} - O_{min}$$

Ideal Straight Line

Definition

For a linear element:

$$O_{ideal} = KI + a$$

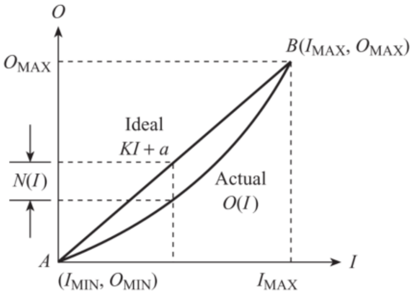
Where:

•K= slope (sensitivity)

•a= intercept (zero offset)

$$K = \frac{O_{max} - O_{min}}{I_{max} - I_{min}}$$

$$a = O_{min} - KI_{min}$$



Exercise

A force sensor has an output range of 1 to 5 V corresponding to an input range of 0 to 2×10^5 N. Find the equation of the ideal straight line.

$$O_{\text{ideal}} = K \cdot I + a$$

$$K = \frac{O_{\text{MAX}} - O_{\text{MIN}}}{I_{\text{MAX}} - I_{\text{MIN}}} \qquad K = \frac{5\ \text{V} - 1\ \text{V}}{2 \times 10^5\ \text{N} - 0\ \text{N}} = \frac{4}{2 \times 10^5}\ \text{V/N} = 2 \times 10^{-5}\ \text{V/N}$$

$$a = O_{min} - KI_{min} \qquad a = 1\ \text{V}$$

$$V = 2 \times 10^{-5} F + 1$$

The unit of force F is Newton (N), and the unit of voltage V is Volt (V).

A differential pressure transmitter has an input range of 0 to 2×10^4 Pa and an output range of 4 to 20 mA. Find the equation to the ideal straight line.

$$O_{\text{ideal}} = K \cdot I + a$$

$$K = \frac{O_{\text{MAX}} - O_{\text{MIN}}}{I_{\text{MAX}} - I_{\text{MIN}}} \qquad K = \frac{20\ \text{mA} - 4\ \text{mA}}{2 \times 10^4\ \text{Pa} - 0\ \text{Pa}} = \frac{16}{20000}\ \text{mA/Pa} = 8 \times 10^{-4}\ \text{mA/Pa}$$

$$a = O_{min} - KI_{min} \qquad a = 4\ mA$$

$$i = 8 \times 10^{-4} \cdot \Delta P + 4$$

Static Characteristics of Measurement System Elements

Non-linearity (Non-linear Error)

1. Why Non-linearity Exists

In practice, most sensors are **not perfectly linear**.

The real input–output relationship does not exactly follow the ideal straight line:

$$O_{ideal} = KI + a$$

Therefore, we define **non-linearity** as the deviation of the actual output from this ideal straight line.

2. Mathematical Definition

If the actual output is $O(I)$,then:

$$N(I) = O(I) - (KI + a)$$

Where:

- $O(I)$ = actual output
- $KI + a$ = ideal straight-line output
- $N(I)$ = non-linearity at input I

3. Non-linearity as Percentage of Full-Scale Deflection (FSD)

Manufacturers usually express non-linearity as a percentage of **span (full-scale deflection)**.

$$\%\ \text{f.s.d.} = \frac{\hat{N}}{O_{max} - O_{min}} \times 100\%$$

Where:

- $\hat{N} = max \mid N(I) \mid$ maximum deviation
- $O_{max} - O_{min}$ = output span

Important: The denominator is the **output span**, not the input span.

Exercise

A non-linear pressure sensor has an input range of 0 to 10 bar and an output range of 0 to 5 V. The output voltage at 4 bar is 2.20 V. Calculate the non-linearity in volts and as a percentage of span.

Final Answer:

1. The non-linearity error of the sensor at 4 bar is **0.20 V**.
2. The non-linearity as a percentage of span is **4%**.

Exercise

A non-linear temperature sensor has an input range of 0 to 400 °C and an output range of 0 to 20 mV. The output signal at 100 °C is 4.5 mV. Find the non-linearity at 100 °C in millivolts and as a percentage of span.

$$N(100^\circ\mathrm{C}) = 4.5\ \mathrm{mV} - 5.0\ \mathrm{mV} = -0.5\ \mathrm{mV}$$

(Note: The magnitude of error usually takes the absolute value, meaning it deviates by ***0.5 mV****.*

In engineering expressions, you can say the maximum non-linearity magnitude is 0.5 mV.)

$$\%\ \mathrm{f.s.d.} = \frac{0.5\ \mathrm{mV}}{20\ \mathrm{mV}} \times 100\% = 2.5\%$$

Exercise

A thermocouple used between 0 and 500 °C has the following input–output characteristics:

Input T °C	0	100	200	300	500
Output E μV	0	5268	10 777	16 325	27 388

(a) Find the equation of the ideal straight line.
(b) Find the non-linearity at 100 °C and 300 °C in μV and as a percentage of f.s.d.

Answer (a):

The ideal straight-line equation for this thermocouple is $E_{\text{ideal}} = 54.776 \times T$ (where the unit of E is μV and the unit of T is °C).

Percentage of f.s.d.:

$$\%\ \text{f.s.d.} = \frac{|N(300)|}{\text{f.s.d.}} \times 100\% = \frac{107.8}{27388} \times 100\% \approx 0.394\%$$

Static Characteristics of Measurement System Elements

Interfering Input (I_I)

Definition: An interfering input is an environmental variable that causes a change in the zero offset (intercept) of the sensor output.

It does not change the slope, but shifts the entire input–output relationship vertically.

Mathematical Expression

For a linear element:

$$O = KI + a$$

If an interfering input I_I exists:

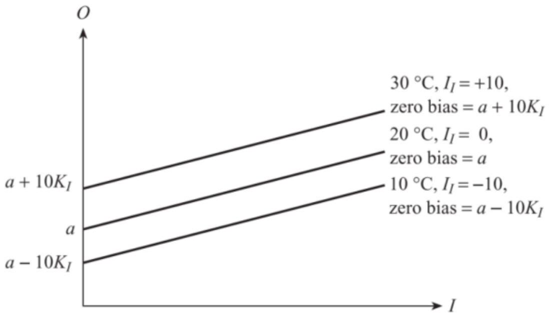
$$O = KI + (a + K_I I_I)$$

Where:

- K_I =environmental sensitivity to interfering input
- I_I =deviation from standard condition

So the intercept changes from a to:

$$a + K_I I_I$$



Static Characteristics of Measurement System Elements

Modifying Input (I_M)

Definition: A modifying input is an environmental variable that changes the sensitivity (slope) of the sensor.

It does not shift the intercept, but changes how steep the input-output curve is.

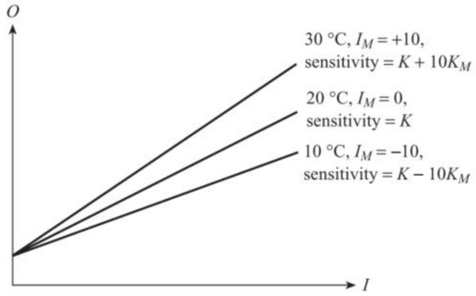
Mathematical Expression

Original linear model:

$$O = KI + a$$

With modifying input:

$$O = (K + K_M I_M)I + a$$



Where:

•K_M = change in sensitivity per unit modifying input

•I_M = deviation from standard condition

So sensitivity becomes:

$$K + K_M I_M$$

Static Characteristics of Measurement System Elements

The Generalized Model of a Measurement Element

Now we combine:

- Ideal linear behavior
- Non-linearity
- Interfering input
- Modifying input

Complete Static Model

$$O = KI + a + N(I) + K_M I_M I + K_I I_I$$

$$O = (K + K_M I_M)I + a + K_I I_I$$

Term	Meaning
$KI + a$	Ideal straight line
$N(I)$	Non-linear deviation (Engineering reason: **N(I) = 0**. To calculate the slope (sensitivity K), it is necessary that this line be a straight line.)
$K_M I_M I$	Gain change due to environment
$K_I I_I$	Zero shift due to environment

Exercise

A force sensor has an input range of 0 to 10 kN and an output range of 0 to 5 V at a standard temperature of 20 °C. At 30 °C the output range is 0 to 5.5 V. Quantify this environmental effect.

Final Answer:

The temperature change for this sensor manifests only as a **modifying input** effect (there is no interfering effect, $K_I = 0$).

This environmental effect can be quantified by the modifying sensitivity constant:

$K_M = 0.005\ \mathrm{V}/(\mathrm{kN} \cdot {}^\circ\mathrm{C})$.

Exercise

A pressure transducer has an output range of 1.0 to 5.0 V at a standard temperature of 20 °C, and an output range of 1.2 to 5.2 V at 30 °C. Quantify this environmental effect.

In this pressure transmitter, the temperature change manifests only as an **Interfering Input** effect (there is no modifying effect, so $K_M = 0$).

This environmental effect can be quantified by the interfering environmental coupling constant:
$K_I = 0.02\ \mathrm{V}/^{\circ}\mathrm{C}$

(Physical meaning: For every $1\ ^{\circ}\mathrm{C}$ *increase in environmental temperature, the output of this transmitter will shift upward as a whole by* $0.02\ \mathrm{V}$*).*

Exercise

A pressure transducer has an input range of 0 to 10^4 Pa and an output range of 4 to 20 mA at a standard ambient temperature of 20 °C. If the ambient temperature is increased to 30 °C, the range changes to 4.2 to 20.8 mA. Find the values of the environmental sensitivities K_I and K_M.

Final Answer:

- The interfering input constant is $K_I = 0.02\ \mathrm{mA}/^{\circ}\mathrm{C}$.
- The modifying input constant is $K_M = 6 \times 10^{-6}\ \mathrm{mA}/(\mathrm{Pa} \cdot {}^{\circ}\mathrm{C})$.

Static Characteristics of Measurement System Elements

ADC (Analog-to-Digital Converter)

An ADC converts a continuous analog signal into a discrete digital code.

- Input: analog voltage V_{in}
- Output: digital binary number

Resolution

Definition:

The smallest change in input voltage that produces a change in output code.

It is also called the **quantization step size**.

Mathematical Expression

For an n-bit ADC:

$$\Delta V = \frac{V_{max} - V_{min}}{2^n} \qquad \%f.s.d. = \frac{1}{2^n} \times 100\%$$

Hysteresis

Definition:

Hysteresis occurs when:

The output depends on the direction of input change.

For the same input value:

- Output differs when input is increasing
- Compared to when input is decreasing

Mathematical representation of hysteresis error:

$$H = V_{up} - V_{down} \qquad \%f.s.d. = \frac{H}{O_{max} - O_{min}} \times 100\%$$

Exercise

An analogue-to-digital converter has an input range of 0 to 5 V. Calculate the resolution error both as a voltage and as a percentage of f.s.d. if the output digital signal is:

(a) 8-bit binary
(b) 16-bit binary.

(a) 8-bit ADC: Resolution error is approximately **19.53 mV**, which is **0.391%** of the span.

(b) 16-bit ADC: Resolution error is approximately **76.29 µV**, which is **0.00153%** of the span.

Exercise

A level transducer has an output range of 0 to 10 V. For a 3 metre level, the output voltage for a falling level is 3.05 V and for a rising level 2.95 V. Find the hysteresis as a percentage of span.

The hysteresis error of this liquid level transmitter is **0.10 V**.

The hysteresis error as a percentage of the span is **1%**.